

Question No:5

DS	Session	Sorting	Question Information	Level 1	Challenge 15
Problem Description: One day Priya was jogging and realized that her life was boring. Everything was red, even the streets in the best park were red. Therefore she decided to make streets a little bit brighter. She know that every street in the park is a segment laying on the X axis with coordinates X_l, X_r ($X_l \leq X_r$). Streets may intersect or overlap. She chooses any subset of streets and paints them in green. After that she wants to get one continuous green segment. As she really likes number L the length of this segment has to be equal to L. Your task is to determine if it is possible to choose some subset of streets and paint them to get one green segment with the length equal to L? Constraints: $1 \leq \text{sum of all } N \leq 2 * 10^3$ $1 \leq L \leq 10^6$ $1 \leq X_l \leq X_r \leq 10^6$ $1 \leq N \leq 20, 1 \leq X_l \leq X_r \leq 200$, holds for test cases worth 10% of the problem's score. $1 \leq N \leq 100, 1 \leq X_l \leq X_r \leq 200$, holds for test cases worth 20% of the problem's score. Input Format: The first line contains one integer T - the number of test cases. Each test case starts with two integers N and L, denoting the number of streets and Priya's favorite number L. The next N lines contain two integers X_l, X_r, denoting the left and right borders of the street. Output Format: Print the output in a separate lines "Yes" if it is possible to paint some street and "No" otherwise.					

Logical Test Cases

Test Case 1

INPUT (STDIN)

2
5 3
1 2
2 3
3 4
1 5
2 6
2 3
1 2
2 6

EXPECTED OUTPUT

1 2
2 6

Test Case 2

INPUT (STDIN)

2
5 3
1 4
2 3
3 5
5 1
5 6
2 3
2 4
1 3

EXPECTED OUTPUT

2 4
1 3

Yes No

Code:

```
1  #include <stdio.h>
2  typedef struct
3  {
4      int l, r;
5  } Street;
6
7  int main()
8  {
9      int T;
10     scanf("%d", &T);
11
12     while (T--)
13     {
14         int N, L;
15         scanf("%d %d", &N, &L);
16
17         Street a[N];
18
19         for (int i = 0; i < N; i++)
20         {
21             scanf("%d %d", &a[i].l, &a[i].r);
22
23             if (a[i].l > a[i].r)
24             {
25                 int temp = a[i].l;
26                 a[i].l = a[i].r;
27                 a[i].r = temp;
28             }
```

```

29     }
30
31     int answer = 0;
32
33     /*
34     * Try every possible left endpoint.
35     */
36     for (int i = 0; i < N && !answer; i++)
37     {
38         int start = a[i].l;
39         int target = start + L;
40         int right = start;
41
42         /*
43         * We can only use streets completely inside
44         * [start, target].
45         */
46         int changed = 1;
47
48         while (changed && right < target)
49         {
50             changed = 0;
51
52             for (int j = 0; j < N; j++)
53             {
54                 /*
55                 * Street must be completely inside

```

```

56                 * the required segment.
57                 */
58                 if (a[j].l >= start &&
59                     a[j].r <= target)
60                 {
61                     /*
62                     * It must touch/overlap the segment
63                     * we have already painted.
64                     */
65                     if (a[j].l <= right && a[j].r > right)
66                     {
67                         right = a[j].r;
68                         changed = 1;
69                     }
70                 }
71             }
72         }
73
74         if (right >= target)
75             answer = 1;
76     }
77
78     if (answer)
79         printf("Yes\n");
80     else
81         printf("No\n");

```

```
82     }  
83  
84     return 0;  
85 }
```

Output:

Test Case-1:

```
PS C:\Users\kesha\OneDrive\Documents\3sem\C>  
2  
5 3  
1 2  
2 3  
3 4  
1 5  
2 6  
Yes  
2 3  
1 2  
2 6  
No
```

Test Case-2:

```
2  
5 3  
1 4  
2 3  
3 5  
5 1  
5 6  
Yes  
2 3  
2 4  
1 3  
Yes  
PS C:\Users\kesha\OneDrive\Documents\3sem\C>
```

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